Estimation of a Real Business Cycle Model via Simulated Method of Moments (SMM)

Macroeconomic Modeling
José Alejandro Rodríguez & Adrián Fernández

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1 Introduction: The Real Business Cycle Model

Historically, macroeconomic analysis relied heavily on the idea that business cycles were the result of market failures or nominal shocks, often viewed through the lens of IS-LM and Phillips Curve models. However, following the Lucas Critique in 1976, which argued that traditional models failed because their parameters shifted in response to policy changes, the macroeconomic paradigm shifted.

The Real Business Cycle (RBC) theory emerged in the 1980s, culminating in Kydland and Prescott's fully articulated Dynamic Stochastic General Equilibrium (DSGE) model. The RBC model proposes a radical departure from previous thought: business cycles are not market failures, but rather efficient, optimal responses by rational agents to exogenous real shocks, primarily in technology and productivity.

1.1 Objective and Core Mechanics

The primary objective of the RBC model is to construct a dynamic framework that fits historical data and can simulate the effects of future policies and shocks. The model relies on a competitive equilibrium where households optimize their utility over an infinite horizon (smoothing consumption) and firms maximize profits under perfect competition. This dynamic is captured by a system of non-linear equations:

- **The Production Function:** $Y_t = A_t K_{t-1}^{\alpha} N_t^{1-\alpha}$, where output is determined by capital, labor, and a stochastic technological component.
- **The Resource Constraint:** $Y_t = C_t + I_t$, ensuring all output is allocated to consumption or investment.
- **The Euler Equation:** $\frac{C_{t+1}}{C_t} = \beta(r_{t+1} + 1 - \delta)$, which dictates the optimal intertemporal path for consumption.
- **The Technology Shock:** $\ln(A_t) = \rho_a \ln(A_{t-1}) + \varepsilon_a$, modeled as an autoregressive process, AR(1), driving business cycle fluctuations.

1.2 The Problem: The Unobservable Engine

To make this model functional, it must be calibrated. While parameters like the depreciation rate (δ) or the capital share (α) can be derived from empirical data, the model's main driver—

the technology shock—presents a significant hurdle. We cannot directly observe aggregate technology in the real world. Critics often refer to this unobservable variable as a measure of ignorance because it captures every fluctuation in output not attributed to capital or labor. To simulate the economy, we need to estimate the magnitude (σ_a) and persistence (ρ_a) of these shocks.

2 Methodological Framework: SMM and GMM

The Generalized Method of Moments (GMM) serves to solve over-identification problems by weighting population moments by their uncertainty. The estimator minimizes a loss function $J(\theta)$ based on the distance between theoretical and sample moments:

$$J(\theta) = g(\theta)' W^{-1} g(\theta) \tag{1}$$

Where W^{-1} is a weighting matrix that assigns less importance to highly volatile moments.

In complex DSGE models where analytical expressions for moments are unavailable, we utilize the Simulated Method of Moments (SMM). By the Law of Large Numbers, the average of thousands of simulations converges to the true theoretical moment. In this assignment, we use SMM to estimate the parameters governing the unobservable technology shock by matching the variance and first-order autocorrelation of the business cycle.

3 Results and Analysis

3.1 Steady-State Equilibrium

The model solved for a deterministic steady state consistent with neoclassical theory. Output stabilizes at $Y = 1.4065$, with Consumption at $C = 1.1151$ and Investment at $I = 0.2913$. The capital stock reached $K = 11.6544$.

3.2 Parameter Estimates and Diagnostic Note

The SMM procedure updated the parameters to fit the targeted moments:

- **Persistence ($\hat{\rho}_a$):** 0.808
- **Volatility ($\hat{\sigma}_a$):** 0.007

Computational Diagnostic: The estimated volatility ($\hat{\sigma}_a$) successfully converged to 0.007, properly identifying the magnitude of the technological shocks without hitting the lower boundary constraint observed in previous iterations.

3.3 Business Cycle Moments and Comovements

The simulated model replicates key stylized facts of the U.S. economy:

- **Consumption Smoothing:** The standard deviation of Consumption (0.0009) is significantly lower than that of Output (0.0029), validating the consumption smoothing hypothesis.

- **Procyclicality:** Consumption and Investment are strongly procyclical, with correlations of 0.733 and 0.955 respectively.
- **Persistence:** The model generates an Output autocorrelation of 0.8039, matching the high persistence observed in empirical U.S. data ($\rho \approx 0.84$).

3.4 Economic Discussion of Impulse Response Functions (IRFs)

The IRFs illustrate the dynamic transmission of a technology shock through the economy based on neoclassical thought:

1. **Intertemporal Substitution of Labor:** A positive productivity shock increases the real wage (w), raising the opportunity cost of leisure. Agents respond by increasing labor supply (N) immediately to take advantage of high productivity.
2. **Consumption Smoothing:** According to the Permanent Income Hypothesis, households do not consume the entire windfall from the shock. Instead, they increase consumption (C) gradually and by a smaller magnitude than output, spreading the gains over time.
3. **Investment Dynamics:** Since the shock is persistent ($\rho_a = 0.808$), the expected future marginal product of capital is high. This triggers a massive surge in Investment (I) to accumulate capital (K), which is the most volatile component of the cycle.
4. **Capital Accumulation:** Higher investment today leads to higher capital tomorrow. This sustains the expansion even as the technology shock begins to dissipate, showing the internal propagation mechanism of the RBC framework.

4 Conclusion

The estimation confirms that a standard RBC model driven by real technological shocks can endogenously replicate fundamental U.S. business cycle characteristics, such as consumption smoothing and high output persistence. The application of SMM allows for the empirical validation of these unobservable drivers, bridging the gap between theoretical DSGE modeling and observed macroeconomic data.